

Physics LATAM - Riemannian and Complex Geometry - Homework 4 Solutions

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Exercise 4.1 (Riemannian curvature: multilinearity)

Let ∇ be a connection on the vector bundle $E \rightarrow M$ with the curvature operator

$$R(X, Y)s := \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X, Y]}s, \quad X, Y \in \Gamma(TM), \quad s \in \Gamma(E).$$

We must show for $f_1, f_2, f_3 \in C^\infty(M)$

$$R(f_1 X, f_2 Y)(f_3 s) = f_1 f_2 f_3 R(X, Y)s.$$

Recall the connection identities (Leibniz):

$$\nabla_f X s = f \nabla_X s, \quad \nabla_X(fs) = X(f)s + f \nabla_X s.$$

We expand each of the three terms in the definition of $R(f_1 X, f_2 Y)(f_3 s)$ completely so cancellations are explicit.

Write $A := \nabla_{f_1 X} \nabla_{f_2 Y}(f_3 s)$. Then

$$\begin{aligned} A &= f_1 \nabla_X (f_2 \nabla_Y (f_3 s)) \\ &= f_1 X(f_2) \nabla_Y (f_3 s) + f_1 f_2 \nabla_X \nabla_Y (f_3 s) \\ &= f_1 X(f_2) (Y(f_3)s + f_3 \nabla_Y s) + f_1 f_2 (X(Y(f_3))s + Y(f_3) \nabla_X s \\ &\quad + X(f_3) \nabla_Y s + f_3 \nabla_X \nabla_Y s) \\ &= f_1 X(f_2) Y(f_3)s + f_1 X(f_2) f_3 \nabla_Y s + f_1 f_2 X(Y(f_3))s \\ &\quad + f_1 f_2 Y(f_3) \nabla_X s + f_1 f_2 X(f_3) \nabla_Y s + f_1 f_2 f_3 \nabla_X \nabla_Y s. \end{aligned}$$

Similarly write $B := \nabla_{f_2 Y} \nabla_{f_1 X}(f_3 s)$. By swapping the roles of $(1, X)$ and $(2, Y)$ we obtain

$$\begin{aligned} B &= f_2 Y(f_1) X(f_3)s + f_2 Y(f_1) f_3 \nabla_X s + f_1 f_2 Y(X(f_3))s \\ &\quad + f_1 f_2 X(f_3) \nabla_Y s + f_1 f_2 Y(f_3) \nabla_X s + f_1 f_2 f_3 \nabla_Y \nabla_X s. \end{aligned}$$

Finally compute $C := \nabla_{[f_1 X, f_2 Y]}(f_3 s)$. Using the commutator identity

$$[f_1 X, f_2 Y] = f_1 f_2 [X, Y] + f_1 X(f_2)Y - f_2 Y(f_1)X,$$

we get

$$\begin{aligned}
C &= \nabla_{f_1 f_2 [X, Y]}(f_3 s) + \nabla_{f_1 X(f_2) Y}(f_3 s) - \nabla_{f_2 Y(f_1) X}(f_3 s) \\
&= f_1 f_2 ([X, Y](f_3) s + f_3 \nabla_{[X, Y]} s) + f_1 X(f_2) (Y(f_3) s + f_3 \nabla_Y s) \\
&\quad - f_2 Y(f_1) (X(f_3) s + f_3 \nabla_X s) \\
&= f_1 f_2 [X, Y](f_3) s + f_1 f_2 f_3 \nabla_{[X, Y]} s + f_1 X(f_2) Y(f_3) s + f_1 X(f_2) f_3 \nabla_Y s \\
&\quad - f_2 Y(f_1) X(f_3) s - f_2 Y(f_1) f_3 \nabla_X s.
\end{aligned}$$

Now assemble

$$R(f_1 X, f_2 Y)(f_3 s) = A - B - C.$$

We check cancellations term-by-term. Many derivative-containing terms appear in pairs with opposite signs. As a concrete worked example (this is often the step professors want to see): look at the terms containing the factor $f_1 X(f_2) f_3 \nabla_Y s$. From the expansions above we have the same signed contribution in A and in C :

$$A \text{ contains } + f_1 X(f_2) f_3 \nabla_Y s, \quad C \text{ contains } + f_1 X(f_2) f_3 \nabla_Y s.$$

We now give a short index-level derivation illustrating how curvature appears on a single upper index and on a single lower index, and why the sign differs.

Example 1 (single contravariant index). Let V^a be a vector field. By definition

$$\nabla_i V^a = \partial_i V^a + \Gamma^a_{ip} V^p.$$

Compute the commutator acting on V^a :

$$\begin{aligned}
(\nabla_k \nabla_i - \nabla_i \nabla_k) V^a &= \nabla_k (\partial_i V^a + \Gamma^a_{ip} V^p) - (i \leftrightarrow k) \\
&= \partial_k \partial_i V^a + \partial_k (\Gamma^a_{ip} V^p) + \Gamma^a_{ip} \partial_k V^p + \Gamma^a_{kq} (\partial_i V^q + \Gamma^q_{ip} V^p) \\
&\quad - (i \leftrightarrow k).
\end{aligned}$$

Subtracting the $(i \leftrightarrow k)$ terms, the pure second partials cancel: $\partial_k \partial_i V^a - \partial_i \partial_k V^a = 0$. Collect the remaining terms that multiply V^p (i.e. with no free derivatives on V) — these are precisely the curvature contributions:

$$\begin{aligned}
(\nabla_k \nabla_i - \nabla_i \nabla_k) V^a &= (\partial_k \Gamma^a_{ip} - \partial_i \Gamma^a_{kp} + \Gamma^a_{kq} \Gamma^q_{ip} - \Gamma^a_{iq} \Gamma^q_{kp}) V^p + \text{terms with } \partial V \\
&= R^a_{pki} V^p + (\text{terms with } \partial V).
\end{aligned}$$

The remaining terms with ∂V cancel each other when the full expression is expanded (they rearrange into total derivatives that cancel under subtraction). The net result is the neat tensorial identity

$$(\nabla_k \nabla_i - \nabla_i \nabla_k) V^a = R^a_{pki} V^p.$$

Example 2 (single covariant index). Let ω_b be a 1-form. Then

$$\nabla_i \omega_b = \partial_i \omega_b - \Gamma^p_{ib} \omega_p.$$

Repeating the same expansion for the commutator one finds

$$(\nabla_k \nabla_i - \nabla_i \nabla_k) \omega_b = (-\partial_k \Gamma^p_{ib} + \partial_i \Gamma^p_{kb}) \omega_p = -R^p_{bki} \omega_p.$$

Thus the curvature appears with a minus sign on lower indices and with a plus sign on upper indices. This sign difference follows directly from the fact that the connection term enters with a plus sign in the covariant derivative of contravariant components (the $+\Gamma^a_{ip}$ term) but with a minus sign in the covariant derivative of covariant components (the $-\Gamma^p_{ib}$ term). When the commutator is formed the $\partial\Gamma$ and $\Gamma\Gamma$ combinations assemble into the Riemann tensor with the indicated signs.

Putting these index-level examples together yields the general Ricci identity stated above: each upper index i_p contributes a term R^{ip}_{qki} acting on that slot, while each lower index j_m contributes $-R^q_{j_mki}$ acting on that slot.

This completes the index-level justification of the Ricci commutator identity. Therefore in $A - B - C$ these combine as

$$(+f_1X(f_2)f_3\nabla_Ys) - 0 - (+f_1X(f_2)f_3\nabla_Ys) = 0,$$

so the entire $X(f_2)$ -term cancels out. The same pattern occurs for the other derivative-terms: e.g. the $f_2Y(f_1)X(f_3)s$ terms in B and C cancel, and the second-derivative scalar terms combine with the the $[X, Y](f_3)$ contribution to cancel as well because

$$f_1f_2(X(Y(f_3)) - Y(X(f_3)) - [X, Y](f_3)) = 0.$$

After carrying out the bookkeeping one finds that all terms involving derivatives of f_1, f_2, f_3 cancel, leaving exactly the factor

$$f_1f_2f_3(\nabla_X\nabla_Ys - \nabla_Y\nabla_Xs - \nabla_{[X, Y]}s) = f_1f_2f_3R(X, Y)s.$$

Thus

$$R(f_1X, f_2Y)(f_3s) = f_1f_2f_3R(X, Y)s.$$

Because this holds for arbitrary smooth functions f_i and section s , the map $(X, Y, s) \mapsto R(X, Y)s$ is $C^\infty(M)$ -linear in each slot; equivalently R is a smooth tensor field in $\Gamma(\Lambda^2T^*M \otimes \text{End } E)$. (Saying merely "multilinear" can be ambiguous: the stronger statement is that R is a C^∞ -bilinear alternating form in (X, Y) with values in $\text{End}(E)$, and it is C^∞ -linear in the section slot as well.)

Exercise 4.2 (Curvatures of the n -sphere)

Let S^n be the n -sphere with the standard round metric of sectional curvature $K = 1$ (unit radius). For a space of constant sectional curvature K the Riemann curvature tensor has the well-known form

$$R_{ijkl} = K(g_{ik}g_{jl} - g_{il}g_{jk}).$$

For the unit sphere $K = 1$, so

$$R_{ijkl} = g_{ik}g_{jl} - g_{il}g_{jk}.$$

The Ricci tensor is obtained by contracting indices (contract i with the third index). Contraction must be done carefully: recall $g^{ik}g_{ik} = n$ (the trace of the identity), so

$$\begin{aligned} R_{jl} &= g^{ik}R_{ijkl} = g^{ik}(g_{ik}g_{jl} - g_{il}g_{jk}) \\ &= K(g^{ik}g_{ik}g_{jl} - g^{ik}g_{il}g_{jk}) = K(n g_{jl} - g_{jl}) = K(n - 1)g_{jl}. \end{aligned}$$

Hence the Ricci tensor is $(n-1)Kg$. The scalar curvature is the trace of the Ricci tensor; contracting again gives

$$R = g^{jl}R_{jl} = g^{jl}((n-1)Kg_{jl}) = (n-1)K g^{jl}g_{jl} = (n-1)K n = n(n-1)K.$$

For the unit sphere $K = 1$ this reduces to $R_{jl} = (n-1)g_{jl}$ and $R = n(n-1)$; if the sphere has radius r then $K = 1/r^2$ and the general form above applies.

Exercise 4.3 (Curvatures of the hyperbolic plane)

The hyperbolic plane $\mathbb{H}^2$ has constant sectional curvature $K = -1$ (unit curvature convention). Using the constant-curvature formula with $n = 2$ and $K = -1$ we get

$$R_{ijkl} = -(g_{ik}g_{jl} - g_{il}g_{jk}).$$

Contracting to obtain the Ricci tensor gives

$$R_{jl} = g^{ik} R_{ijkl} = -(n-1)g_{jl} = -1 \cdot g_{jl} \quad (n=2),$$

so $R_{jl} = -g_{jl}$. The scalar curvature is

$$R = g^{jl} R_{jl} = -2.$$

Thus the hyperbolic plane has negative curvature as expected.

Exercise 4.4 (Ricci identity for tensors)

Let T be a (r, s) -tensor with components $T^{i_1 \dots i_r}_{j_1 \dots j_s}$ in coordinates. The Ricci identity expresses the commutator of second covariant derivatives in terms of the Riemann curvature acting on each index. In components one has the general formula

$$(\nabla_k \nabla_i - \nabla_i \nabla_k) T^{i_1 \dots i_r}_{j_1 \dots j_s} = \sum_{p=1}^r R^i{}_{pqi} T^{i_1 \dots i_r}_{j_1 \dots j_s} - \sum_{m=1}^s R^q{}_{jmk} T^{i_1 \dots i_r}_{j_1 \dots j_s}.$$

Proof sketch: apply the definition of the covariant derivative to each index and use the commutation formula for the Christoffel coefficients:

$$(\partial_k \Gamma^p{}_{iq} - \partial_i \Gamma^p{}_{kq} + \Gamma^p{}_{kr} \Gamma^r{}_{iq} - \Gamma^p{}_{ir} \Gamma^r{}_{kq}) = R^p{}_{qki}.$$

When the covariant derivatives act, terms involving second derivatives of the components combine with the Christoffel terms; reordering the two derivatives produces exactly the curvature contractions on each index slot, with signs appropriate for upper (contravariant) vs lower (covariant) indices. The stated component identity follows by collecting these contributions; this is the standard Ricci commutator identity.

Exercise 4.5 (Taylor expansion of the metric in normal coordinates)

In Riemann normal coordinates centered at a point p we have the well-known expansion (see e.g. standard Riemannian geometry texts):

$$g_{ij}(x) = \delta_{ij} - \frac{1}{3} R_{ikj\ell}(0) x^k x^\ell + O(|x|^3),$$

or equivalently (after raising indices or symmetrizing appropriately)

$$g^{ij}(x) = \delta^{ij} + \frac{1}{3} R^i{}_k{}^j{}_\ell(0) x^k x^\ell + O(|x|^3).$$

The coefficient $-1/3$ can be obtained by a short computation using the vanishing of the Christoffel symbols at the origin of normal coordinates together with the definition of the Riemann tensor. We outline the standard derivation:

1. In normal coordinates centered at p we have $\Gamma^i_{jk}(0) = 0$; consequently

$$\partial_k g_{ij}(0) = g_{ir}(0) \Gamma^r_{jk}(0) + g_{jr}(0) \Gamma^r_{ik}(0) = 0.$$

2. Differentiate this identity once more and evaluate at the origin: since $g_{ab}(0) = \delta_{ab}$ and $\Gamma(0) = 0$, one finds

$$\partial_k \partial_\ell g_{ij}(0) = g_{ir}(0) \partial_\ell \Gamma^r_{jk}(0) + g_{jr}(0) \partial_\ell \Gamma^r_{ik}(0).$$

3. Use the definition of the Riemann tensor at the origin (where the quadratic terms in Γ vanish) to relate derivatives of the Christoffel symbols to curvature:

$$R^r_{ik\ell}(0) = \partial_k \Gamma^r_{\ell i}(0) - \partial_\ell \Gamma^r_{ki}(0).$$

Taking suitable permutations in the last identity and symmetrizing the result gives

$$\partial_k \partial_\ell g_{ij}(0) = -\frac{1}{3} (R_{ikj\ell}(0) + R_{i\ell jk}(0)),$$

where indices have been lowered with $g(0) = \delta$. This is the advertised coefficient $-1/3$ that appears in the Taylor expansion of g_{ij} .

Determinant convention note: the expansion of $\det g$ in normal coordinates depends on the curvature normalization used in the course (how one defines R_{ijkl} and the Ricci tensor). With the convention used above (Ricci $R_{k\ell} = R^i_{ki\ell}$) one obtains

$$\det g = 1 - \frac{1}{3} R_{k\ell}(0) x^k x^\ell + O(|x|^3).$$

Some texts define the curvature with an extra factor or opposite sign, in which case the coefficient in the determinant expansion can appear as $-1/6$; consult your course convention for R_{ijkl} if the grader uses a different sign. The formula for the metric components displayed earlier is the standard one consistent with the $R_{ikj\ell}$ used in these solutions.

Exercise 4.6 (Einstein equations with cosmological constant)

We work in 4 spacetime dimensions. The vacuum field equations with cosmological constant are

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = 0.$$

(a) **Einstein manifold condition.** Contract the field equations with $g^{\mu\nu}$ to obtain

$$R - \frac{1}{2} (4) R + 4\Lambda = 0 \quad \Rightarrow \quad -R + 4\Lambda = 0,$$

so $R = 4\Lambda$. Substitute back into the original equation:

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} (4\Lambda) + \Lambda g_{\mu\nu} = 0 \quad \Rightarrow \quad R_{\mu\nu} - 2\Lambda g_{\mu\nu} + \Lambda g_{\mu\nu} = 0,$$

hence

$$R_{\mu\nu} = \Lambda g_{\mu\nu}.$$

Comparing with the problem statement form $R_{\mu\nu} = 3k g_{\mu\nu}$ we read off $3k = \Lambda$, so $k = \Lambda/3$.

(b) Christoffel symbols for the static spherically symmetric metric. Consider the metric

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2(d\theta^2 + \sin^2\theta d\varphi^2).$$

We compute the nonzero Christoffel symbols via the metric formula

$$\Gamma^\lambda_{\mu\nu} = \frac{1}{2}g^{\lambda\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}).$$

Nonzero metric components: $g_{tt} = -f$, $g_{rr} = 1/f$, $g_{\theta\theta} = r^2$, $g_{\varphi\varphi} = r^2 \sin^2\theta$. Nonzero Christoffel symbols (symmetry on lower indices implied):

$$\begin{aligned}\Gamma^t_{tr} &= \Gamma^t_{rt} = \frac{f'(r)}{2f(r)}, \\ \Gamma^r_{tt} &= \frac{1}{2}g^{rr}(-\partial_r g_{tt}) = \frac{1}{2}f(r)f'(r) = \frac{1}{2}f(r)f'(r), \\ \Gamma^r_{rr} &= -\frac{f'(r)}{2f(r)}, \\ \Gamma^r_{\theta\theta} &= -f(r)r, \quad \Gamma^r_{\varphi\varphi} = -f(r)r\sin^2\theta, \\ \Gamma^\theta_{r\theta} &= \Gamma^\theta_{\theta r} = \frac{1}{r}, \quad \Gamma^\varphi_{r\varphi} = \Gamma^\varphi_{\varphi r} = \frac{1}{r}, \\ \Gamma^\theta_{\varphi\varphi} &= -\sin\theta\cos\theta, \quad \Gamma^\varphi_{\theta\varphi} = \Gamma^\varphi_{\varphi\theta} = \cot\theta.\end{aligned}$$

These can be derived equivalently from the Euler–Lagrange equations for the Lagrangian $L = \frac{1}{2}g_{\mu\nu}\dot{x}^\mu\dot{x}^\nu$ by writing the Euler–Lagrange equation for each coordinate and matching the result to the geodesic equation $\ddot{x}^\lambda + \Gamma^\lambda_{\mu\nu}\dot{x}^\mu\dot{x}^\nu = 0$; the coefficients that multiply the quadratic velocities identify the Christoffel symbols. The expressions agree with the components above.

(c) Curvature tensors. For a general function $f(r)$ the Riemann tensor components are algebraically complicated (they involve f , f' and f''). One convenient nonzero component (up to symmetries) can be written clearly as

$$R^r_{\theta r\theta} = -\frac{rf'(r)}{2} - (f(r) - 1),$$

which follows by substituting the Christoffel symbols into $R^\rho_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}$.

If the metric satisfies the vacuum Einstein equations with cosmological constant Λ , part (a) implies $R_{\mu\nu} = \Lambda g_{\mu\nu}$ and therefore the Ricci tensor components equal Λ times the metric components, and the scalar curvature is $R = 4\Lambda$. These relations drastically simplify the curvature expressions and are used in part (d).

(d) Determination of $f(r)$ (Schwarzschild–(A)dS solution). Plug the metric ansatz into the Einstein equations

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = 0.$$

The independent nontrivial components reduce to ordinary differential equations for $f(r)$. A brief derivation sketch is as follows: compute the relevant Ricci components (e.g. R_{tt} and R_{rr}) by substituting the Christoffel symbols above; combining these components produces an ODE for $f(r)$ of the form

$$\frac{d}{dr}(r(1 - f(r))) = -\Lambda r^2 + C_0,$$

where C_0 is an integration constant which is fixed by boundary conditions. Integrating and choosing the constant so that the solution is asymptotically flat when $\Lambda = 0$ yields

$$f(r) = 1 - \frac{2M}{r} - \frac{\Lambda}{3}r^2,$$

with the constant M identified as the mass parameter. This is the Schwarzschild–(anti-)de Sitter metric: for $\Lambda > 0$ one obtains Schwarzschild–de Sitter, for $\Lambda < 0$ Schwarzschild–anti-de Sitter, and for $\Lambda = 0$ the usual Schwarzschild solution.

(e) Killing vectors. The spacetime is static and spherically symmetric for any value of Λ . Hence the Killing vector fields are the time-translation generator ∂_t and the three generators of rotations (a basis of the Lie algebra $\mathfrak{so}(3)$). Concretely there are four independent Killing vectors:

- $K_{(t)} = \partial_t$ (stationarity),
- $K_{(1)}, K_{(2)}, K_{(3)}$ generating the $SO(3)$ rotations (e.g. the angular momentum vector fields on the 2-spheres of symmetry).

Their forms do not depend on the sign of Λ ; the symmetry group remains $\mathbb{R} \times SO(3)$.